The empirical recurrence for OEIS A217340, proved by transfer matrix

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A217340 counts the distinct OCCUPANCY arrays obtained when every cell of an $n \times 2$ grid holds one token and each token moves to a neighbouring cell, and carries an empirical recurrence of order 4 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. What is counted is the size of an IMAGE — two different assignments of moves with the same occupancy count once — so it is not a walk count in the obvious graph; determinising the machine that emits the occupancy row by row turns it into one, on $S = 28$ states, after which the conjectured recurrence is settled by a finite exact computation.

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1 The conjecture

OEIS A217340 is "Number of n X 2 arrays of occupancy after each element moves to some horizontal or vertical neighbor but without 2-loops.". It has offset 1 and begins

$$0, 1, 7, 38, 190, 918, 4367, 20623, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: a(n) = 7*a(n-1) -11*a(n-2) +a(n-3) -a(n-4).

As of the "Last modified" line on the live entry (Jul 22 2025 23:39:20, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

Every cell of an $n \times 2$ grid holds one token. Each token moves by one of the offsets

$$(-1, 0), (0, -1), (0, 1), (1, 0),$$

a move being available only when it lands inside the grid. The offset $(0, 0)$ is NOT among them: every token must move. The OCCUPANCY array records, for each cell, how many tokens land on it; the entry counts how many different occupancy arrays occur as the assignment of moves runs over all its possibilities. The entry also forbids 2-loops: no two tokens may exchange places.

That is the size of an IMAGE, and it is the whole difficulty: two assignments with the same occupancy must be counted once, and a walk count over the assignments would count them twice.

3 The image is a walk count after determinisation

Every offset moves the row index by at most one, so the occupancy of row i is decided by the moves chosen in rows $i-1$, i and $i+1$. Writing $\beta(p,c,x)$ for that occupancy row, with p or x allowed to be the symbol $\perp$ for "outside", the whole occupancy array is

$$\bigl(\beta(\perp,c_1,c_2),\ \beta(c_1,c_2,c_3),\ \ldots,\ \beta(c_{n-1},c_n,\perp)\bigr)$$

for an assignment $c_1,\ldots,c_n$ of move-rows. The symbol $\perp$ is not decoration: whether "up" or "down" is a legal move for a cell depends on it, and β returns nothing at all when the row it is given cannot occur in that position.

Determinise. For a set D of pairs (p,c) and an occupancy row b put

$$\delta(D,b)=\{(c,x):(p,c)\in D,\ \beta(p,c,x)=b\},\qquad f(D)=\bigl|\{\beta(p,c,\perp):(p,c)\in D,\ \beta(p,c,\perp)\text{ defined}\}\bigr|,$$

and let $D_0=\{(\perp,c):c\}$. An induction on the number of rows shows D_t is exactly the set of pairs of consecutive move-rows completing a prefix with the emitted occupancy rows, so that with M the adjacency matrix of the digraph on the reachable nonempty sets, one edge per occupancy row it admits,

$$a(n)\;=\;\iota^\top M^{n-1}\tau,\qquad \iota=e_{D_0},\quad \tau=f.$$

There are $S=28$ reachable sets. In particular a is C-finite.

4 The proof

Lemma 1. *Let $q(t)=t^4-\sum_i c_i t^{4-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n)-\sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau=0$ for all large j, and it is enough to examine $j<S$.*

Proof. Factor M^j out of $M^{j+4}-\sum_i c_i M^{j+4-i}$. For the bound, M^S is an integer combination of $I,M,\ldots,M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j=\iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n)\;=\;7\,a(n-1)-11\,a(n-2)+a(n-3)-a(n-4)$$

for every $n>4$.

Proof. The vector $w=q(M)\tau$ was formed by 4 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j=0,1,\ldots,$ again exactly; those integers vanish from the index corresponding to $n=4+1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 23 terms the entry publishes, exactly, in integer arithmetic. The entry's first term is 0, and it is a sharp test of the boundary: a one-row grid offers only the moves that stay within one row, so a neighbour set with no horizontal offset gives no legal assignment at all and the count is zero, which is what the entries of that shape publish.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A217340.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.